

B.3 Spherical Coordinates

We first choose an origin. Then we choose a coordinate, r , that measures the radial distance from the origin to the point P . The coordinate r ranges in value from $0 \leq r < \infty$. The set of points that have constant value for r are spheres (“level surfaces”).

Any point on the sphere can be defined by two angles (θ, ϕ) and r . We will define these angles with respect to a choice of Cartesian coordinates (x, y, z) . The angle θ is defined to be the angle between the positive z -axis and the ray from the origin to the point P . Note that the values of θ only range from $0 \leq \theta \leq \pi$. The angle ϕ is defined (in a similar fashion to polar coordinates) as the angle in the between the positive x -axis and the projection in the x - y plane of the ray from the origin to the point P . The coordinate angle ϕ can take on values from $0 \leq \phi < 2\pi$.

The spherical coordinates (r, θ, ϕ) for the point P are shown in Figure B.3.1. We choose the unit vectors $(\hat{r}, \hat{\theta}, \hat{\phi})$ at the point P as follows. Let $\hat{r}$ point radially away from the origin, and $\hat{\theta}$ point tangent to a circle in the positive θ direction in the plane formed by the z -axis and the ray from the origin to the point P . Note that $\hat{\theta}$ points in the direction of increasing θ . We choose $\hat{\phi}$ to point in the direction of increasing ϕ . This unit vector points tangent to a circle in the xy -plane centered on the z -axis. These unit vectors are also shown in Figure B.3.1.

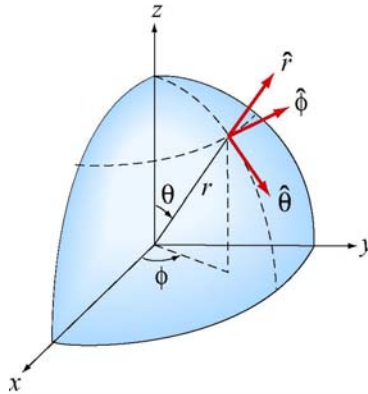


Figure B.3.1 Spherical coordinates

If you are given spherical coordinates (r, θ, ϕ) of a point in the plane, the Cartesian coordinates (x, y, z) can be determined from the coordinate transformations

$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned} \tag{B.3.1}$$

Conversely, if you are given the Cartesian coordinates (x, y, z) , the spherical coordinates (r, θ, ϕ) can be determined from the coordinate transformations

$$\begin{aligned} r &= +(x^2 + y^2 + z^2)^{1/2} \\ \theta &= \cos^{-1} \left(\frac{z}{(x^2 + y^2 + z^2)^{1/2}} \right) \\ \phi &= \tan^{-1}(y/x) \end{aligned} \quad (\text{B.3.2})$$

The unit vectors also are related by the coordinate transformations

$$\begin{aligned} \hat{\mathbf{r}} &= \sin \theta \cos \phi \hat{\mathbf{i}} + \sin \theta \sin \phi \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}} \\ \hat{\boldsymbol{\theta}} &= \cos \theta \cos \phi \hat{\mathbf{i}} + \cos \theta \sin \phi \hat{\mathbf{j}} - \sin \theta \hat{\mathbf{k}} \\ \hat{\boldsymbol{\phi}} &= -\sin \phi \hat{\mathbf{i}} + \cos \phi \hat{\mathbf{j}} \end{aligned} \quad (\text{B.3.3})$$

These results can be understood by considering the projection of $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}})$ into the unit vectors $(\hat{\boldsymbol{\rho}}, \hat{\mathbf{k}})$, where $\hat{\boldsymbol{\rho}}$ is the unit vector from cylindrical coordinates (Figure B.3.2),

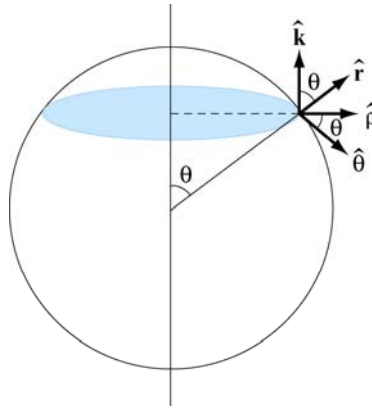


Figure B.3.2 Cylindrical and spherical unit vectors

$$\begin{aligned} \hat{\mathbf{r}} &= \sin \theta \hat{\boldsymbol{\rho}} + \cos \theta \hat{\mathbf{k}} \\ \hat{\boldsymbol{\theta}} &= \cos \theta \hat{\boldsymbol{\rho}} - \sin \theta \hat{\mathbf{k}} \end{aligned} \quad (\text{B.3.4})$$

We can use the vector decomposition of $\hat{\boldsymbol{\rho}}$ into the Cartesian unit vectors $(\hat{\mathbf{i}}, \hat{\mathbf{j}})$:

$$\hat{\boldsymbol{\rho}} = \cos \phi \hat{\mathbf{i}} + \sin \phi \hat{\mathbf{j}} \quad (\text{B.3.5})$$

To find the inverse transformations we can use the fact that

$$\hat{\rho} = \sin \theta \hat{r} + \cos \theta \hat{\theta} \quad (\text{B.3.6})$$

to express

$$\begin{aligned} \hat{i} &= \cos \phi \hat{\rho} - \sin \phi \hat{\phi} \\ \hat{j} &= \sin \phi \hat{\rho} + \cos \phi \hat{\phi} \end{aligned} \quad (\text{B.3.7})$$

as

$$\begin{aligned} \hat{i} &= \cos \phi \sin \theta \hat{r} + \cos \phi \cos \theta \hat{\theta} - \sin \phi \hat{\phi} \\ \hat{j} &= \sin \phi \sin \theta \hat{r} + \sin \phi \cos \theta \hat{\theta} + \cos \phi \hat{\phi} \end{aligned} \quad (\text{B.3.8})$$

The unit vector $\hat{k}$ can be decomposed directly into $(\hat{r}, \hat{\theta})$ with the result that

$$\hat{k} = \cos \theta \hat{r} - \sin \theta \hat{\theta} \quad (\text{B.3.9})$$

B.3.1 Infinitesimal Line Element

Consider a small infinitesimal displacement $d\vec{s}$ between two points (Figure B.3.3). This vector can be decomposed into

$$d\vec{s} = dr\hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi} \quad (\text{B.3.10})$$

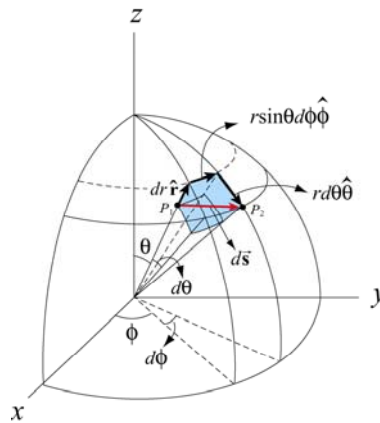


Figure B.3.3 Infinitesimal displacement vector $d\vec{s}$.

B.3.2 Infinitesimal Area Element

Consider an infinitesimal area element on the surface of a sphere of radius r (Figure B.3.4).

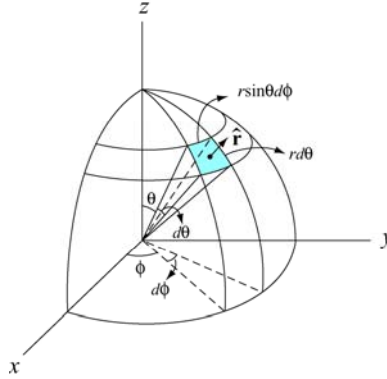


Figure B.3.4 Area element for a sphere.

The area of this element has magnitude

$$dA = (r d\theta)(r \sin \theta d\phi) = r^2 \sin \theta d\theta d\phi \quad (\text{B.3.11})$$

points in the radially direction (outward from the surface of the sphere). So for the surface of the sphere, the infinitesimal area vector is

$$d\vec{A} = r^2 \sin \theta d\theta d\phi \hat{r} \quad (\text{B.3.12})$$

B.3.3 Infinitesimal Volume Element

An infinitesimal volume element (Figure B.3.5) is given by

$$dV = r^2 \sin \theta d\theta d\phi dr \quad (\text{B.3.13})$$

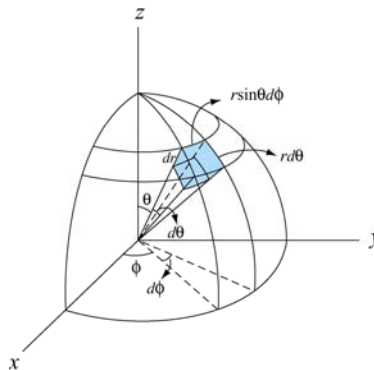


Figure B.3.5 Infinitesimal volume element.